MZUZU UNIVERSITY

Department of Mathematics

EMT 1101 - Precalculus

End of Semester Examination

Tuesday, April 4, 2017 **Time:** 8 : 00 – 11 : 00 **(3 hours)**

1. Check that the paper has 13 questions.
2. Attempt all questions
3. Write answers in the space provided
4. Show all your working clearly
5. Use proper notation
6. Simplify your answers as far as possible
7. No Calculators

Name _____

Reg Number , Programme: _____

Lecturer's name: _____

1. (a) Use properties of exponents to simplify

i. $\left(\frac{x^2y^{20}}{x^6y^{15}}\right)^2$, (2)

ii. $(3^{1+\sqrt{5}})(3^{2-\sqrt{5}})$. (2)

(b) Evaluate $\sqrt[3]{-\frac{8}{125}}$. (2)

(c) Rationalize the denominator and simplify (3)

$$\frac{1}{\sqrt{5}} + 4\sqrt{45}.$$

2. (a) Determine whether the following pair of equations form lines that are parallel, perpendicular or neither; (3)

$$5x - 6y = 25; 6x + 5y = 0$$

(b) Find the equation of the line that passes through $P(1, 3)$ and is perpendicular to the line passing through points $(-2, -3)$ and $(6, -1)$. (4)

3. Consider a circle with equation

$$x^2 + y^2 - 2x - 6y - 10 = 0.$$

(a) Find the radius and the coordinates of the center. (4)

(b) Verify that the point $(3, 7)$ lies on the circle. (2)

4. Solve the following inequality presenting your answer using interval notation (4)

$$\left|\frac{x+1}{2} - \frac{(x-1)}{3}\right| < 1.$$

5. (a) Let $f(x) = 2x + 3$. Find values for a and b such that the equation $f(ax + b) = x$ is true for all values of x. (3)

(b) Let $k(x) = 5x^3 + \frac{5}{x^3} - x - \frac{1}{x}$. Show that $k(x) = k\left(\frac{1}{x}\right)$. (2)

6. (a) Determine the domain of $h(x) = \sqrt{-2x^2 - x + 6}$ and present it using interval notation. (6)

(b) Determine the range of $h(x) = \frac{x^3+3}{x^3-5}$ and present it using interval notation. (3)

7. (a) Let $f(x) = x^2$ and $g(x) = 2x - 1$. (5)
Compute $\frac{f[g(x)]-f[g(a)]}{g(x)-g(a)}$

(b) Let $f(x) = 4x^2$. Find $\frac{f(x)-f(-2)}{x+2}$ (3)

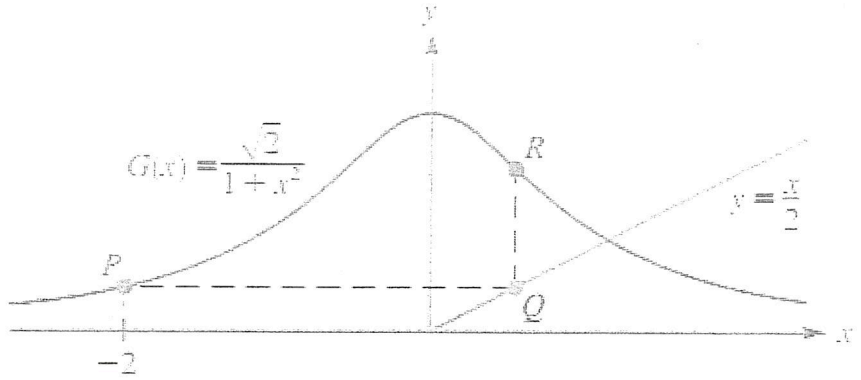
8. (a) Solve the equation (3)

$$5 = 2e^{2x-1}$$

and express your answer in terms of natural logarithms.

(b) Find the domain of $y = \log_{10}(x^2 - 25)$. (6)

9. Determine the coordinates of the points P,Q,and R. Assume that each dashed line is parallel to one of the coordinate axes. (6)



10. Use translations to sketch the graph of $y = \frac{1}{x+2} + 2$, specifying intercepts and asymptotes. (6)

11. Show that (5)

$$\cot\theta + \frac{1 - 2\cos^2\theta}{\sin\theta\cos\theta} = \tan\theta.$$

12. (a) Convert $\frac{\pi}{12}$ radians to degrees. (2)

(b) Convert 660° to radians. (2)

13. (a) If $\sin t = -\frac{2}{3}$ and $\pi < t < \frac{3\pi}{2}$, compute $\cos t$ and $\tan t$. (4)

(b) Evaluate $\csc\left(-\frac{3\pi}{4}\right)$ (4)